

**IMPACT OF NUTRIENT-ENRICHED BIOCHAR ON SOIL PROPERTIES
AND THE YIELD FORMATION OF FABA BEAN (*VICIA FABA* L.)**

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ABSTRACT

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Soil water and nutrient, as well as climate change are affecting the global agricultural practices. As a soil amendment, softwood biochar is a carbon-rich substances produced from the pyrolysis of biomass has already become an interesting research topic. But the nutrient-enriched biochar impact has lack of scientific literature and field experiment. So, this study explores the impact of nutrient-enriched biochar on the yield formation of faba bean (<i>Vicia faba</i> L.), with a particular focus on soil moisture content and soil chemical properties. The research aimed to assess how varying application rates of biochar and NPK fertilizers influence soil moisture retention, electrical conductivity, pH, nutrient availability, total concentration of carbon, nitrogen, the overall growth and yield formation of faba bean. The experimental field setup was done at the Viikki Experimental Farm, University of Helsinki, Finland, where different biochar treatments (0 ton/ha, 5+15 ton/ha, and 10 ton/ha) were combined with varying fertilizer application rates (30%, 65%, and 100% of the recommended dose). The results indicated that nutrient-enriched biochar (5+15 ton/ha) significantly ($p < 0.05$) improved soil moisture content on the surface (0-18 cm) soil, particularly during critical growing periods, and enhanced vegetative biomass by 105.84%, leaves weight by 173.53%, above ground biomass by 73.44% of faba bean compared to no biochar treatment. Total concentration of carbon also increased by 32.83%. Fertilizer application also significantly ($p < 0.001$ mostly) increased soil nutrient contents, and interactions between biochar and fertilizer had minimal effects on soil properties. The findings highlight the potential of biochar as a sustainable soil amendment that can improve the resilience and biomass production of faba bean, particularly in water- deficient regions and can improve the soil carbon storage capacity.			
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List of abbreviations

AGB	Above Ground Biomass
B	Boron
BC	Biochar
BC × F	Biochar × Fertilizer
C	Carbon
Ca	Calcium
CEC	Cation Exchange Capacity
Cu	Copper
EC	Electrical Conductivity
F	Fertilizer
HI	Harvest Index
K	Potassium
Mg	Magnesium
Mn	Manganese
P	Phosphorous
PWP	Permanent Wilting Point
S	Sulfur
TDR	Time Domain Reflectometry
Zn	Zinc

1. Introduction

Climate change presents several challenges to global agriculture, affecting water availability, crop yields, and agricultural productivity (Prajapati et al., 2024). The main challenge in modern agricultural systems is increasing crop production while ensuring sustainability and minimizing environmental impact (Hamilton et al., 2016). Biochar is observed as a viable option for addressing problems related to food security for the growing population of the world (Tilman et al., 2011) and is being investigated for its advantages worldwide (Shackley et al., 2012; Nair et al., 2017). The application of several types of biochar can be an important tool for achieving agricultural sustainability by reducing greenhouse gas emissions and improving soil fertility (Gogoi et al., 2019).

In recent years, biochar produced through the pyrolysis of organic substances like wood and manure (Lehmann & Joseph, 2015) has attracted significant research and application interest, as reported through scientometric analysis and reviews by Wu et al. (2019). When different types of biochar are incorporated into soil, depending on the compound and chemical properties of the biochar, the carbon (C) decomposition rates vary (Kuzyakov et al., 2014), and biochar can remain in the soil for hundreds to thousands of years (Lu et al., 2020).

Numerous studies have been conducted on different kinds of biochar. Biochar's ability to boost yields is particularly relevant in tropical and subtropical areas (Jeffery et al., 2017; Ye et al., 2020). In tropical soils, biochar can release essential plant nutrients for growth (Alling et al., 2014), enhancing soil fertility, particularly by boosting soil organic carbon, cation exchange capacity (CEC), and the availability of phosphorus, potassium, calcium, and magnesium, which can lead to higher yields (Sukartono et al., 2011). In the humid tropics, biochar influences the microbial population by modifying the soil environment and enhancing labile active carbon sources (Simarani et al., 2018). In field experiments in temperate soils, biochar can increase water holding capacity, with improvements or reductions in crop yield depending on the application rate and the type of soil (Karer et al., 2013).

Though biochar is typically used without any treatment, recent studies indicate that it may be enhanced through physical or chemical modification to improve its performance (Fakhar

et al., 2025). The effect of nutrient-enriched biochar has been studied (Domingues et al., 2017; Eazhilkrishna et al., 2017; Schmidt et al., 2017; Wali et al., 2022) in tropical regions for different types of crops. Biochar produced from softwood biomass influences soil physical properties, such as soil moisture content and crop biomass yield, depending on the rate of biochar application, type of soil, and the crops over different years since application (Tammeorg, 2014a; Kalu et al., 2021). Biochar derived from softwood biomass can also affect soil chemical properties, nutrient uptake, and plant growth (Yi et al., 2019). However, the effect of nutrient-enriched softwood biochar on soil properties and yield components of faba bean (*Vicia faba* L.) has not been studied at the field level, according to our search of previous research findings. Faba bean is a widely consumed grain legume utilized as both human food and animal feed due to its superior nutritional qualities as a pulse (Cazzato et al., 2012). Faba beans are also valued for their ecological contributions, such as nitrogen fixation, which improves soil fertility and reduces the need for synthetic fertilizers, in addition to their nutritional advantages (Rfaki et al., 2018).

The impact of nutrient-enriched softwood biochar in agricultural fields needs to be explored for soil properties and several yield components in temperate soils. Therefore, this study aims to investigate the impact of nutrient-enriched biochar on soil moisture content, soil chemical properties, and yield formation in faba bean. This study is part of a wider research project conducted by the Agrichar Research Group at the University of Helsinki, Finland.

2. Literature review

2.1 Biochar

According to the European Biochar Certificate (EBC), biochar is a porous, carbon-rich substance generated through the pyrolysis of biomass, utilized to sequester carbon as a carbon sink or to substitute fossil carbon in industrial production, and is not intended for combustion to produce energy. A thermal process is required, in which organic materials are decomposed at temperatures between 350°C and 1000°C in an environment with limited to no oxygen (EBC, 2012-2024). The EBC characterizes biochar by its quality traits, the type of raw materials used in its production, the sustainability of the production process, and its intended applications.

Biomass is converted into carbon-rich, micro-porous materials with a high degree of aromatization and a well-developed porous structure through pyrolysis (Lehmann & Joseph, 2015). Due to its aromatic structure, biochar is highly resistant to microbial degradation, which enhances its ability to contribute to soil carbon sequestration (Brown et al., 2020). The production of biochar is dependent on several components, particularly the type of biomass, its moisture content, particle size, and reaction factors such as time, temperature, and heating rates (Kumar et al., 2020; Ge et al., 2021).

The biomass type and pyrolysis temperature have an impact on the cation exchange capacity (CEC) of biochar (Suliman et al., 2016). As the temperature rises, biochar aromatizes and becomes more hydrophobic, reducing the number of functional groups holding oxygen and nitrogen, while decreasing its ability to hold cations like calcium (Ca^{2+}), magnesium (Mg^{2+}), potassium (K^+), sodium (Na^+), and water (Shinogia & Kanri, 2003; Moreno-Castilla, 2004; Kinney et al., 2012). The oxygen/carbon (O/C) ratio and thus CEC of biochar may rise as it ages because some functional groups on its surface may undergo oxidation processes, producing more oxygen-containing functional groups (Clough & Condon, 2010; Atkinson et al., 2010). The specific surface area of biochar typically ranges from 1.5 to 500 m^2/g (Suliman et al., 2016; Li et al., 2018) and increases with rising pyrolysis temperatures within a certain range, such as from 200°C to 800°C (Al-Wabel et al., 2013). The temperature increase from 300°C to 500°C can result in surface area increasing from 1.20 m^2/g to 393 m^2/g (Handiso et al., 2024).

2.2 Characteristics of softwood biochars

If biochar is produced under the same pyrolysis conditions, biochars made from wood typically exhibit a higher surface area and lower ash content compared to those derived from grass biomass due to the varying properties of the biomass, such as the levels of cellulose, hemicellulose, lignin, and mineral ash present (Han et al., 2013; Ronsse et al., 2013). The Brunauer, Emmett, and Teller (BET) surface area for pinewood biochar is 219.35 m²/g (Jiang et al., 2017). Biochar materials derived from softwood significantly decrease NH₃ volatilization, making it available for nitrification, which turns it into NO₃⁻ (Heaney et al., 2018).

According to the review study by Sohi et al. (2010), as pyrolysis temperature increases, the oxygen content in biochar decreases, resulting in softwood biochar produced at higher temperatures typically containing a higher amount of recalcitrant organic carbon, such as aromatic carbon. This carbon structure provides the biochar with significant chemical and biological stability. Additionally, softwood biochar, like biochar in general, has a low density and contains various functional groups, including hydroxyl, nitro, ester, carboxyl, and carbonyl groups (Sohi et al., 2010).

Softwood biochar typically has a higher carbon content, greater specific surface area, and higher volatile matter compared to biochar produced from other materials and animal dung feedstocks, which exhibit a lower cation exchange capacity. However, it is more porous and has a higher carbon/nitrogen ratio, making it suitable for enhancing soil structure and providing long-term carbon sequestration benefits (He et al., 2024).

2.3 Biochars as soil amendments

One of the first examples of incorporating biochar into the soil environment is the Terra Preta in the southern Amazon Basin (Glaser et al., 2001). Wim Sombroek, a Dutch soil expert, discovered black matter-rich soil in Brazil's Amazon basin in the 1960s. This soil, now known as Terra Preta, or the black soil of the Indians, had a very high concentration of organic matter, nitrogen (N), phosphorus (P), potassium (K), and other plant nutritional components (Harder, 2006; Marris, 2006; Tenenbaum, 2009).

Archaeologists discovered by examining the soil's constituents that it contains bone fragments from various animals, including fish, crop residues, ceramic carbon residue, and human-burned wood (Sombroek et al., 2002). Its pore structure makes the soil more fertile and conducive to plant growth by facilitating the accumulation of nutrients and advantageous microbes (Cheng et al., 2008).

Softwood biochars can be utilized as a soil amendment since they are mostly made up of stable and recalcitrant organic carbon (C) (Schmidt et al., 2002; Xiang et al., 2017). In the soil, newly produced biochar undergoes oxidation through biotic and abiotic processes, enhancing the negative charge on its surface and promoting nutrient retention capacity (Chen, 2008). Biochar as an amendment increases P and K availability in the soil, leading to an increase in P and K concentrations in the leaves at the growing stage of plants (Piccolo et al., 2022). Biochar addition also reduces the mobilization of heavy metals like cadmium (Cd) from soil to the root and shoot in rice cultivation (Han et al., 2024).

2.4 Biochar and soil properties

The effects of applying biochar on soil properties have been the subject of numerous studies in recent decades (Lehmann et al., 2011; Chen et al., 2008; Ameloot et al., 2013; Zhu et al., 2017). The physical and chemical characteristics of soil, such as pH, nutritional status, cation-exchange capacity (CEC), and moisture content, have generally improved with the use of biochar (Lehmann, 2007; Jeffery et al., 2011; Noguera et al., 2012; Wang et al., 2015; Xiang et al., 2017).

Soil moisture is an important factor influencing plant growth and productivity. Biochar application can reduce the rate of evaporation in the cultivated layer and improve the water holding capacity of soil in the field (Feng et al., 2023). Biochar application can improve soil water retention capacity in several ways, including reducing soil bulk density (Abel et al., 2013), increasing total average pore size (Obia et al., 2016), and increasing soil surface area (Laird et al., 2010). The raw material used to make biochar can affect how well it retains water. For instance, biochar made from woody biomass typically retains moisture better than biochar made from agricultural waste (e.g., rice husks, wheat straw, cotton stalks, or crop residues) because it has a higher surface area and porosity (Spokas et al., 2012). According to the meta-analysis of Razzaghi et al. (2020), the addition of biochar led to a 51% increase

in field capacity for coarse-textured soils. In contrast, field capacity for medium-textured soils showed a modest rise of 13%, while fine-textured soils showed almost no change, with field capacity remaining less than 1% after biochar application. The application of softwood biochar (pyrolyzed at 600–700°C) to finer-textured soil resulted in a 217% increase in the mean weight diameter of aggregates, indicating enhanced soil aggregation, and this increase in aggregation was associated with a significant rise in physically protected carbon stored within macro aggregates (Wang et al., 2017).

The addition of biochar substantially reduced bulk density and enhanced soil porosity (Singh et al., 2022), as porosity and bulk density are negatively correlated (Kakaire et al., 2015). The use of biochar significantly increased the diversity of bacteria and fungi, with the labile carbon and micro/macronutrients it provides to microbial communities in soils being responsible for this enhancement (Sohi et al., 2010; Xu et al., 2016). Previous studies have consistently shown that biochar materials enhance soil nutrient availability in highly weathered tropical soils (Glaser et al., 2002; Lehmann et al., 2002, 2003; Rondon et al., 2007). Major et al. (2010) found that biochar application increased calcium (Ca) and magnesium (Mg) content by up to 320% and 217%, respectively, on a maize field where the biochar application rate was 20 tons/ha.

2.5 Biochar and yield formation

The impact of biochar on crops yields varies depending on experimental settings such as soil type and application rate (Liu et al., 2013). Research has shown that using biochar can increase crop yields and biomass production (Glaser et al., 2022; Jeffery et al., 2011; Spokas et al., 2012; Biederman & Harpole, 2013). Arif et al., (2017) found that treating soil with biochar before cultivation of winter wheat increased above and below-ground biomass by 81% relative to the untreated control. Biochar application increased crop yield at the 3Mg/ha rate of biochar application (Crane-Droesch et al., 2013). On average, biochar application has been observed to boost biomass of agricultural crops by 10% to 30% (Biederman et al., 2013; Crane-Droesch et al., 2013; Jeffery et al., 2011; Liu et al., 2013).

Rice yields increased by 8.8% to 14%, according to Zhang et al. (2010). Double rice output has a significantly greater harvest index (Qin et al., 2016) that indicates the ratio of grain yield to the total biomass. However, certain combinations of soil and biochar have also been

shown to reduce agricultural productivity (Van Zwieten et al., 2010; Wisnubroto et al., 2010). For example, Asai et al. (2009) found that applying biochar reduced grain production by 10% to 23.3%. According to a review of the biochar literature by Spokas et al. (2012), 20% of the studies showed adverse effects on growth or yield, and 30% of the studies found no significant differences between biochar application and control treatment.

Applying biochar tends to provide more favorable growth conditions, especially in soils that are acidic or coarse-textured (Spokas et al., 2012). Due to soluble P from biochar and P retention by biochar (Mukherjee & Zimmerman., 2013) nutrient sorption (Park et al., 2011; Lashari et al., 2013) and soil liming effects (Major et al., 2010) strong positive plant responses to biochar have also been seen in acidic infertile tropical soils. Biochar does not always show the consistence increase of crop biomass. A research by Kalu et al., (2021) on Stagnosol where softwood biochar rate was 30 Mg/ha and the recommended fertilizer level was 30% and 100%, the above ground biomass (AGB) did not show any significant result from the consecutive years (2010 to 2017) on different types of crop. There, only the exception was in case of barley in 2013 and peas in 2016 when biochar addition and biochar × fertilizer addition was significant.

2.6 Biochar effect on yield components of legume

Some research has indicated that stronger growth responses occurred in legumes than in non-legume crops (Thomas & Gale., 2015; Liu et al., 2013). According to Lehman et al. (2006), using biochar increased crop productivity by 20% to 120%. After using biochar, maize grain production can be greatly increased by 98% to 150% (Uzoma et al., 2011). A study by Bangash et al. (2023) on biochar application at 50 tons ha⁻¹ enhanced wheat yield components like tiller/m² by 3%, spikes/m² 6.5%, grains/spike by 3.7%, 1000 grain weight by 1.8% and grain yield by 7.8% . In a 2 year (2018 and 2019) field experiment of cow-pea in a water deficit soil, application of biochar 8ton/ha increased biomass production by 10% in first year and 8% in second year compared to control treatment (Moosavi et al., 2020). Biochar application at 50ton/ha improved the yield of maize by 22% and wheat by 4% (Arif et al., 2015). A study on mung bean by Jalal et al. (2024) applied biochar on 4 different level (0, 10, 20 and 30 ton/h) revealed that 30ton/ha application rate showed the maximum amount of pods/plant (27.3), grains/pods (11.4), grain yield (1550 kg ha⁻¹ and biological

yield (6497 kg/ha) but 1000 grain weight (64 g) was recorded highest weight under 20 ton/ha.

2.7 Biochar and faba bean

Faba bean (*Vicia faba* L.) a leguminous crop also known as fava bean, broad bean, and horse bean, is among the oldest crops grown in the world. (Mínguez & Rubiales., 2021). Faba bean contains high levels of complex carbohydrates, dietary fiber, lysine-rich protein, non-nutrient secondary metabolites, and bioactive substances (phenols, γ all of which have been linked to a number of health advantages (Khazaei et al., 2019; Liu et al., 2022). Among cool-season legumes, faba beans have the highest capacity for effective nitrogen fixation (Mekkei, 2014). Despite its adaptability to various climates, the species' low and inconsistent yields limit its competitiveness as a crop (Duc et al., 2015). Rondon et al. (2007) and Mia et al. (2014) suggested that biochar could enhance legume growth and yield by promoting biological nitrogen fixation.

Suppadit et al. (2012) observed significant increases in plant height, dry weight, yield, and nutrient content in certain legumes following biochar application. Study by Muhamed et al. (2017) found the dry weight of various faba bean parts was significantly influenced by biochar dose up to 80 g per pot resulted in the highest improvements in dry weights due to enhanced nodulation, nitrogen fixation, and nutrient availability following biochar application. Biochar at a rate of 20ton/667 m² increased the fresh pod length of faba bean and increase of yield at a rate of 30 ton/667 m² (Liu et al., 2024). It was also found that biochar effect on faba bean yields components like vegetative mass (g/m²), seeds/plant, seed yield (ton/ha) and 1000 seed weight (g) might not always be significant in different rate of biochar application (Tammeorg et al., 2014a).

2.8 Softwood biochar effect on boreal soil and temperate climate

An experiment on Finnish boreal soil conducted by Tammeorg et al. (2014a) observed that softwood biochar application to fertile sandy clay loam soil was affected several soil properties, including moisture content and nutrient availability. In their study, three application rates (0, 5, and 10 tons/hectare) were tested over three years. The results showed that the higher biochar application rate (10 t dry matter (DM)/ ha) significantly

increased soil carbon content, particularly in the first year, while the 5 t DM/ha rate had a less noticeable effect (Tammeorg et al., 2014a). In the same study, biochar did not show significant improvement of crop yields in any of the tested years, although it did show a positive effect on certain yield components, such as seed number per plant in faba bean and turnip rape.

Karer et al. (2013) observed on temperate soil that biochar significantly increased the soil pH in Cambisol and improved water-holding capacity, especially under drought conditions where the water holding capacity of Cambisol was enhanced by up to 21% after biochar application at a rate of 72 t ha⁻¹. These improvements were less pronounced in Chernozem but still contributed to increased plant-available water. This results align with studies by Karhu et al. (2011) and Jeffery et al. (2011), who found that biochar can increase soil moisture retention, which is crucial for plant growth in regions experiencing periodic droughts. Kalu et al. (2021) examined the impacts of softwood biochar on two contrasting boreal soils: a nutrient-poor Umbrisol and a fertile Stagnosol, over eight years and observed increased plant potassium (K) content, particularly in Umbrisol, while it reduced the uptake of sodium (Na) and aluminum (Al) in Stagnosol. Softwood biochar significantly increased the availability of potassium (K) in the soil while nitrogen (N) uptake by wheat was minimum in the study of Tammeorg et al. (2014b).

Biochars derived from softwood possess the ability to hold mineral nitrogen for prolonged durations (Peltokangas, K. 2023). Softwood biochar may not affect the abundance of earthworms as well as biomass after 6month of biochar application in the field experiment (Tammeorg et al., 2014c). A study of softwood biochar on Finnish soil showed that during extremely dry growing season, softwood biochar amended soil can reduce the nitrous oxide emission but was able to make taller plant than the kontrol treatment but no other significant change in crop biomass nor even shoot-root ratio (Kulmala et al., 2022).

Though softwood biochar effect has a few research on soil properties and yield formation of several legumes, the nutrients enriched softwood biochar effect on soil properties and it's effect on yield components of faba bean is a new research area to explore in field level of experiment.

3. Research objectives

The primary aim of this study was to explore the effect of organic nitrogen loaded biochar (0, 5+15 and 10 ton/ha), NPK fertilizer (30%, 65% and 100% of the recommended level) and their interaction on yield formation of faba bean (*Vicia faba* L.).

The specific objectives of the study were to determine the effect of nutrient-enriched biochar, fertilizer and their interaction on:

- I. Soil moisture content
- II. Soil chemical properties like electrical conductivity, pH and available nutrients (Ca, P, K, Mg, S, B, Cu, Mn, Zn)
- III. The development of yield components and overall yield formation of faba bean

The hypothesis of the study is as follows:

“The application of organic nitrogen-loaded biochar, in combination with varying levels of NPK fertilizer, will significantly affect soil moisture content, soil chemical properties (such as electrical conductivity, pH, and nutrient availability), and the development of yield components, leading to enhanced yield formation of faba bean (*Vicia faba* L.).

4. Materials and Methods

4.1. Study area

The field experiment was carried out at the Viikki Experimental Farm, University of Helsinki, Finland. The experimental site was the Porvoontienvarsi field (Figure 1) which coordinated at 60°13'28.4"N 25°01'36.7"E. The soil was fertile Luvic Stagnosol with a sandy clay loam texture (WRB, 2007). The pH of the soil was 6.6 (Kalu, 2022). The field had a long-term record of softwood biochar treatments since May 2010 with several types of crops (Table 1).



Figure 1: Aerial view of Porvoontienvarsi field, collected from Google Maps, 2024 (<https://maps.google.com/>).

Table 1. The research site's cropping history on the field since 2010.

Year	Grown crops
2010	Wheat (<i>Triticum aestivum</i> L.)
2011	Turnip rape (<i>Brassica rapa</i> L.)
2012	Faba bean (<i>Vicia faba</i> L.),
2013	Barley (<i>Hordeum vulgare</i> L.)
2014-2015	Timothy grass (<i>Phleum pratense</i> L.) with red clover (<i>Trifolium pratense</i> L.)
2016	Oats (<i>Avena sativa</i> L.)
2017	Peas (<i>Pisum sativum</i> L.)
2018	Barley (<i>Hordeum vulgare</i> L.),
2019	Grasses, mostly timothy (<i>Phleum intermedium</i> Jord.) undersown with Barley (<i>Hordeum vulgare</i> L.)
2020-2022	Combination of grasses, mainly timothy (<i>Phleum pratense</i> L.) red clover (<i>Trifolium pratense</i> L.)
2023	Barley (<i>Hordeum vulgare</i> L.)
2024	Faba bean (<i>Vicia faba</i> L.)

4.2 Biochar activation

The biochar used in this study was sourced from Carbofex Oy, Tampere, Finland. The biochar was produced by continuously pyrolyzing Spruce (*Picea abies* L.) wood chips at 550 °C for 10 to 15 minutes, followed by moisture level 20% using water spray. In spring 2023, biochar activation was done with 60% nitric acid (HNO₃), diluted with 975 liter tap water into a 1000 litre IBC container to make the final concentration 0.337M and pH was 0.5. The target was to make 2.5 ml concentrated acid per liter of biochar. Biochar (100 litre) was evenly placed at the IBC (Intermediate Bulk Container) containers, then 10 liter of acid solution with watering was added. Then another 100liter of biochar was added and this process was repeated until the 1000 litre IBC container filled with acid treated biochar and left for a few hours to make it stable. After stabilizing, the acid treated biochar was further added with a slurry fertilizer mixture. The organic fertilizer solution consisted of cattle slurry from the Viikki farm's manure separator, mixed with Agra 13-0-0 (N-P-K) organic fertilizer pellets. The organic nitrogen solution was applied in successive layers onto the previously acid activated biochar using a non-nozzle watering can. It was then left undisturbed by overnight. The next day, the activated biochar was gently bagged for further use on the top layer of the experimental field to prepare the field for the growing of faba bean in 2024.

4.3 Experimental field design

The main plot factor used in the initial biochar application in 2010 was the rate of biochar (0, 5, and 10 ton/ha), and the subplot factor was the rate of fertilizer application in percentage, which was 30, 65, and 100 based on the level recommended for each different type of crop (Tammeorg et al., 2014a; Kalu et al., 2021). In 2024, the study followed the same split-plot design in 4 replicates (I , II , III , IV) as the initial study (Figure 2). Each plot was $2.2 \times 10 \text{ m}^2$ and buffer plots of same size were placed at the beginning and end of each of the three plots. In the spring of 2023, organic nitrogen-loaded activated biochar was applied at a rate of 15 ton/ha in those plot that was 5 ton/ha in 2010. So, the new experiment was considering 5+15 ton/ha nutrient-enriched biochar. There was a 10m corridor between the replicates (Figure 2). After applying the biohar on the experimental field, on 14th of May 2024, faba bean (cultivar name: 'Vire') seeds were sown and the harvesting was done completely on 30th August 2024. The crop was fertilized with mixture of mineral fertilizers so that the 100% rate covered fully the soil-test-based nutrient requirements to faba bean according to local recommendations (Table 2).

Table 2: The nutrient elements of mineral fertilizer treatments for faba bean

Fertilizer rate (%)	Applied easily soluble nutrients in 2024, kg/ha								
	N	P	K	S	Mg	B	Cu	Mn	Zn
100	43.86	15.00	50.00	38.25	9.03	0.60	0.14	10.00	0.00
65	26.61	9.10	33.14	24.86	5.87	0.39	0.09	6.07	0.00
30	12.28	4.20	19.14	11.47	2.71	0.18	0.04	2.80	0.00

Abbreviations: N= nitrogen, P= phosphorous, K= potassium, S= sulphur, Mg= magnesium, B= boron, Cu= copper, Mn= manganese, Zn= zinc

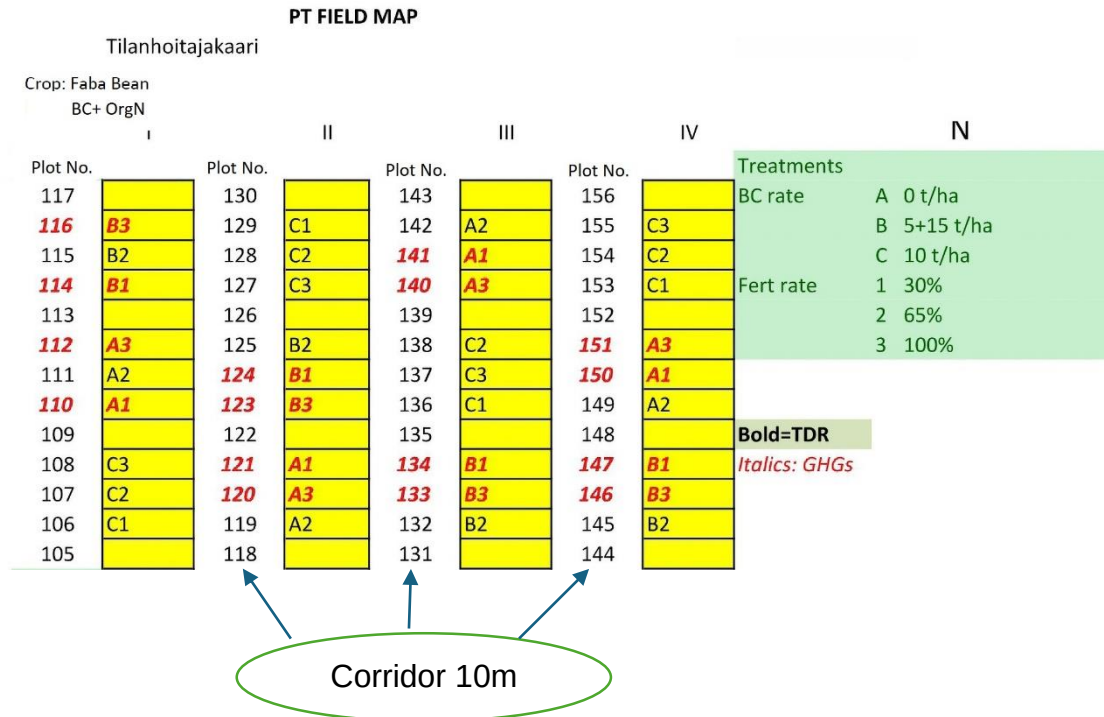


Figure 2. Map of the experimental site showing the replicates, corridors, and biochar and fertilizer rates done in 2023.

4.4 Weather condition and sampling for 2024

In 2024, the growing season was warmer than the previous three years growing seasons as well as 30 years (1991-2020) average monthly temperature (table 2). July and August showed the highest mean temperature in the growing season. May had the lowest amount of mean precipitation (3.2 mm), and September had the highest amount of precipitation when harvesting was already completed. In comparison to 30 years mean monthly precipitation, the growing season had less rainfall from May to August. The growing season had a record number of warmer days than earlier (FMI. 2024).

Table 3: Monthly mean temperature (°C) and precipitation (mm) data throughout the growing season from 2021 to 2024 and the 30-year long-term average data (1991–2020). Data was collected from the Finnish Meteorological Institute, Kumpula weather station, Helsinki, Finland (<https://en.ilmatieteenlaitos.fi/>).

Months	Mean monthly air temperature (°C)					Mean monthly precipitation (mm)				
	1991-2020	2021	2022	2023	2024	1991-2020	2021	2022	2023	2024
May	9.7	10.5	9.8	11.2	14.4	37.0	86.5	50.7	18.0	3.2
June	14.4	19.7	17.2	17.4	17.7	59.0	38.7	39.7	2.8	26.0
July	17.7	21.6	18.6	17.1	19.3	56.0	47.3	48.5	82.4	55.9
August	16.8	15.9	19.4	17.8	18.3	79.0	132.4	45.6	166.1	70.0
September	12.3	10.1	10.3	15.5	15.3	56.0	42.8	64.3	69.7	100.0

Several activities related to agriculture (Table 4), like field preparation, growth stage monitoring, plant and soil samples collection were done throughout the growing season.

Table 4. Agricultural activities conducted during the experimental year 2024; date and description of the activity.

Activity	Date	Description
Tillage	14.05.2024	All plots tilled to 10 cm with 1.2 m wide horizontal power harrow
Fertilization	15.05.2024	2 m Juko machine to about 3-4 cm depth
Sowing	15.05.2024	Wintersteiger 1.25 m wide plot sowing machine
Soil moisture content	23.05.24-11.09.2024 (once in a week)	Time-domain reflectometer (TDR) machine, depth of 0-18cm from the soil surface
Plant development stage	25.05.2024-28.08.24 (once in a week)	Eye observation by comparing to BBCH-scale (Meier. 2018)
Herbicide application	06.06.2024	Basagran SG 1.5 kg/ha, Super Targa 2.0 l/ha + Sito Plus 0.2 l/ha, water level 200 l/ha
Chlorophyll content (SPAD)	26.06.2024	20 readings per plot: SPAD-502 chlorophyll meter ((Minolta Camera Co. Ltd., Chou-Ku, Osaka, Japan)

Leaf Area Index (LAI)	26.06.2024 & 23.07.2024	SunScan SS1 device (Delta-T Devices Ltd, Cambridge, United Kingdom)
Plant sampling	27.08.2024	3 samples/each plot, 50 cm of sowing row
Harvesting	30.08.2024	Combine harvester but used separately for each plot
Soil sampling	10.10.2024	Upto 20cm depth with augers

4.5 Soil Moisture Content Measurement

In the experimental field, no extra irrigation was provided during the study period, natural rainfall being the only source of water. This setup aimed to reflect normal field conditions and evaluate how treatments performed under natural moisture levels. During the growing season, soil moisture content was monitored weekly using Time Domain Reflectometry (TDR; HandiTrase, 6085, Santa Barbara, USA). Soil moisture was monitored at 0-18cm depth from soil surface until 2 weeks after harvesting (Figure 3). The moisture content was measured by TDR as volumetric water content (%) directly.

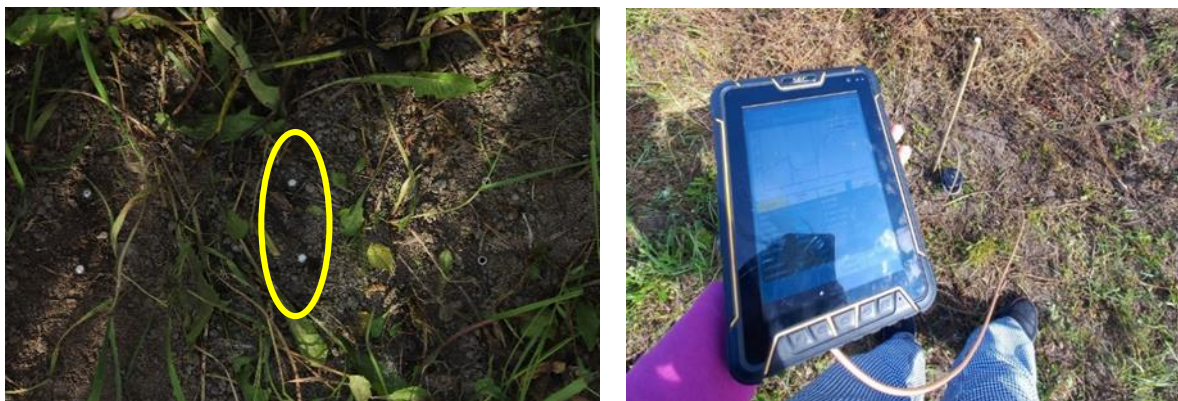


Figure 3. Soil moisture content measuring TDR probes (left image) and using TDR in the field after harvesting (right image).

4.6 Yield Component Analysis

Aboveground biomass (AGB) sampling was conducted at the end of the growing season when all pods had fully matured and dried when the plant development stage was more than 91 (Meier. 2018) (Figure 4). Plant samples were collected as three replicate samples per plot, from three different places of each sowing rows. Within the allotted space of each plot, a 50 cm marker was positioned in a row chosen at random. Every faba bean plant in this 50 cm section was properly uprooted. Then the plant samples were put in a paper bag plot wise and placed into an oven for 72 hours at 60 °C to dry.



Figure 4. Observation of plant development stage before harvesting.

After drying, the yield components sorting was done according to Tammeorg et al. (2014a) where the quantity of plants in each sample was noted, and the plants were separated (Figure 5) into two categories: reproductive organs like pods and vegetative mass (leaves + stems). All leaves that were greater than 2 mm in diameter were taken into account during weighting. The seeds were separated from the pods manually. By sieving and carefully checking the dust and small plant particles, the seeds were counted in the seed counting machine (Pfeuffer Contador Seed Counter, Kitzingen, Germany) 2 times to reduce the error in case of counting. The total weight of the seeds was recorded through the weighing scale. Plants per m² and vegetative mass per m² were calculated, representing the equivalents of the number of plants and the weight of the vegetative mass per m², respectively. After the samples were threshed, the harvest index (HI) was computed, and the weight and quantity of seeds were noted.



Sample before sorting

Stems

Pods

Leaves

Figure 5. Yield components sorting of faba bean.

Leaf area index (LAI) was assessed twice during the flowering stage using the SunScan SS1 device (Delta-T Devices Ltd, Cambridge, United Kingdom). For each plot, three individual measurements were obtained, and the mean of these values was calculated to represent the LAI for each plot.

The relative leaf chlorophyll content was measured in two ways; using a SPAD-502 portable chlorophyll meter (Minolta Camera Co. Ltd., Chou-Ku, Osaka, Japan) and the absolute chlorophyll measurement was done with MC-100 Chlorophyll Concentration Meter (Apogee Instruments, Logan, Utah, USA). For each plot, a total of 20 readings were taken, with 10 readings collected from each side of the plot. The average of these readings was calculated to provide a single representative SPAD value for the entire plot.

4.7 Soil sampling and determination of soil chemical properties

Soil samples were taken about a month after the harvest from a depth of 20 cm using a regular soil auger (Figure: 6) which was marked up to 20cm depth. Total 12 (8+4) sub-samples were taken to make a composite soil sample. Samples were collected from each plot: two from each corner, ensuring a distance of at least 50 cm from the plot boundaries, and four from the midpoint. All of those samples were mixed in a bucket to prepare one soil sample per plot, and larger debris was removed from sample. Then the sample placed into a bag and stored in a cold room before further analysis. The soil samples were sent to commercial laboratory Hortilab OY for further chemical analysis.

Soil-water-suspension 1:2.5 was used to measure the soil electrical conductivity. Potentiometric determination in soil-water-suspension 1:2.5 was used to measure the soil pH. Available nutrients concentration (mg/kg) of calcium (Ca), phosphorus (P), potassium (K), magnesium (Mg), sulfur (S), boron (B), copper (Cu), manganese (Mn) and zinc (Zn) were determined by acid ammonium acetate extraction on a volumetric basis (Vuorinen & Mäkitie.,1955) method then the extracted samples were followed by the elemental analysis of the extracts by inductively coupled plasma optical emission spectrometry (ICP-OES; Thermo Elemental: iCAP 6500 Duo View, Thermo Fisher Scientific, Cambridge, UK) but P was measured with a photometrical analysis (Skalar Analytical Bluvion: SAN++ classic CFA, type 2SAN59000, Skalar, Breda, the Netherlands).



Figure 6: Soil sampling to determine soil chemical properties (left) and Soil sample preparation for total C and N analysis (right)

For determining total carbon (C) and nitrogen (N) concentration, soil samples were dried at oven at maximum 40°C for 48 hours. Then soil was grinded with mortar and pestle (Figure: 6) and 0.18g soil was taken to measure C and N. The concentrations of total nitrogen and carbon were measured using the Dumas technique with an LECO TruMac CN analyzer (LECO Corporation, St. Joseph, MI, USA).

4.8 Statistical analysis

The experimental design involved a two-way split-plot analysis of variance (ANOVA), where biochar application rates and fertilizer levels, along with their interactions, were set as the fixed factors in the model. The design allowed for the analysis of the main effects of biochar and fertilizer, as well as the interaction between these two factors, on various response variables, including soil moisture, nutrient availability, and yield components. The two-way split-plot ANOVA was conducted to examine the effects of different addition levels of biochar and fertilizer on soil moisture content (0-18 cm depth). Biochar treatments were considered as the main plot factor, while fertilizer treatments were assigned as the subplot factor within each main plot. To check the normal distribution of data residuals, Shapiro-Wilk test and detrended normal Q-Q plot was observed. When the model residuals were not normally distributed, box-cox (Box & Cox, 1964) transformation was conducted.

To assess significant differences between treatment means, the Tukey's Honest Significant Difference (HSD) test was applied for multiple pairwise comparisons. The post-hoc test was used after ANOVA to determine which specific treatment means are significantly different from each other. The threshold for statistical significance was set at $p < 0.05$, meaning that any observed differences between treatments with a p-value less than 0.05 were considered statistically significant and $p < 0.001$ was highly statistically significant.

The results of the ANOVA were supplemented with partial eta squared (η^2) values to evaluate the effect sizes of each factor (biochar, fertilizer, and their interaction) on the response variables. Effect sizes greater than 0.14 were considered large, indicating that biochar and fertilizer treatments had a substantial influence on the soil properties and yield components measured in the study. Statistical analysis of the data from the experiments was performed using the software package SPSS version 27.0 (SPSS Corp., Chicago, USA).

5. Results

5.1. Soil moisture content determination

Addition of nutrient enriched biochar increased the moisture content of top (0-18cm) soil layer (Figure 7). The application of biochar at rates of 5+15 tons per hectare resulted highly statistically significant increase in soil moisture content compared to the control treatment (no biochar) across multiple weeks, specifically during the 2nd, 3rd, 4th and 14th weeks after sowing. Additionally, a statistically significant improvement in soil moisture was observed at week 16 during the growing period and also after post-harvest that was at week 17 and 18 after sowing (Figure 7).

From the 7th to the 12th week after sowing, when maximum temperature conditions were recorded, the plots without biochar consistently remained below the permanent wilting point (PWP) throughout the observation period. On the other hand, the plots amended with the highest levels of biochar (5+15 t/ha) exhibited a distinct amount of soil moisture content compared to no biochar treatment. However, the maximum biochar containing plot soil moisture level dropped slightly below the PWP threshold was observed at the 7th week after sowing and the highest decline of soil moisture below the PWP was exhibited at the 12th week after sowing.

Weekly precipitation varied during the whole growing season and sometimes there was no precipitation (2nd, 7th, 11th and 17th weeks after sowing) at all. The weekly average temperature ranged from 15°C to 22°C, with the highest temperature occurring in 3rd, 7th and 11th weeks after sowing which was the stage before seed filling. During the maximum temperature experiencing weeks, all the no biochar containing plots remained below the PWP. Important phenological stages such as flowering occurred in between weeks 7 and 8 after sowing, seed filling during week 14 after sowing date, and crop harvesting was done during in between weeks 16 and 17 after sowing (Figure 7).

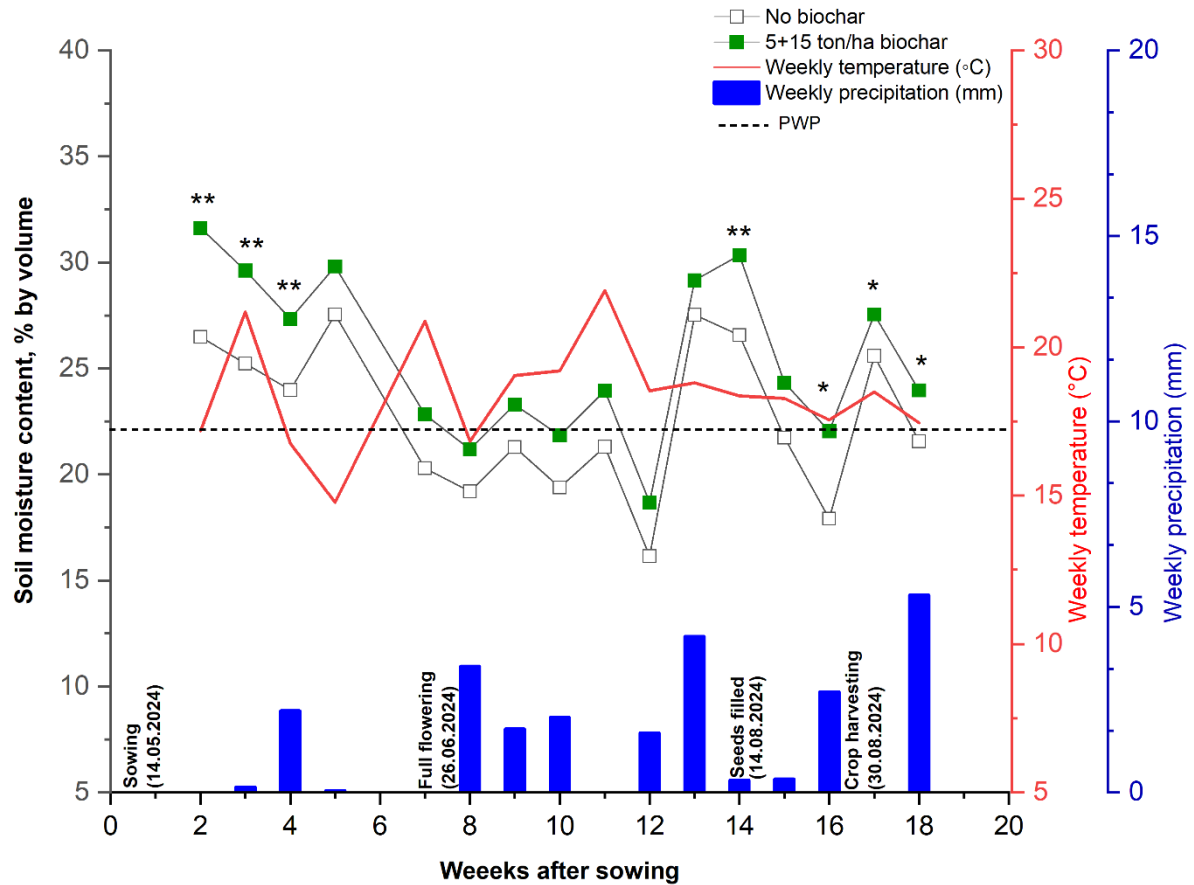


Figure 7: The volumetric water content of the top layer (0–18 cm) in control (no biochar) treatment and in 5+15 ton/ha biochar treatment, along with the weekly average precipitation (mm) and air temperature (°C), were monitored throughout the 2024 growing season. PWP refers to the permanent wilting point moisture content. PWP corresponding to a soil water potential of 1,500 kPa (Tammeorg et al., 2014a) , was determined through laboratory analysis following pressure plate method by Dane and Hopmans (2002). Statistically significant differences between no biochar treatment and 5+15 ton/ha biochar treatment were marked with asterisks; significant levels *, ** were $p < 0.05$ and 0.01 , respectively.

5.2 Determination of soil chemical properties

Electrical Conductivity (EC) and pH of the soil were not significantly affected by biochar rate (BC), fertilizer rate (F), or their interaction (BC×F) (Table 4). However, partial η^2 values indicated a large effect of biochar and interaction of biochar and fertilizer on EC. Additionally,

biochar ($\eta^2 = 0.546$), fertilizer ($\eta^2 = 0.210$) and interaction of biochar and fertilizer ($\eta^2 = 0.165$) showed large effect on soil pH.

Biochar application had only significance impact on the total concentration of soil carbon ($p=0.003$), otherwise biochar did not show any significance result with the concentration of any determined nutrients from soil.

Fertilizer application rate had a significant impact on concentrations of plant available phosphorus (P), potassium (K), magnesium (Mg), sulfur (S), boron (B), manganese (Mn) and zinc (Zn) (Table 3). Copper (Cu) was exceptional as it did not show any significant relation with application of fertilizer. Phosphorus, potassium and sulphur levels increased significantly from minimum fertilizer level (F30) to maximum fertilizer level (F100). Large effect sizes were shown by partial η^2 values, which showed that fertilizer had strong effects on calcium, phosphorus, potassium, magnesium, sulfur, boron, manganese and zinc especially on P ($\eta^2 = 0.891$), S ($\eta^2 = 0.810$), and K ($\eta^2 = 0.804$) except copper which showed small effect size on fertilizer application. Larger effect size also found in case of biochar application where P, K, Mg, S, Cu, Mn, Zn, N and C. The effect size of biochar application was small in Ca and B.

The biochar rate 10 ton/ha tended to increase soil Mg concentration compared to control but no other significant effects on their nutrient concentrations by the addition of biochar were observed (Table 3). Nutrient enriched biochar (5+15 ton/ha) increased soil total carbon concentration significantly which is 32.83% more than the control treatment.

Manganese and zinc concentrations increased notably with higher fertilization, especially at F100. Fertilizer had a large effect on Mn and Zn. The interaction effects (B×F) were significant only for boron ($p = 0.029$), with a large effect size ($\eta^2 = 0.434$) (Table 4).

Table 4: Soil electrical conductivity (EC), pH and plant available nutrient concentrations in soil under different treatments. Values represent the means of 4 replicates for each of the 3 levels of treatment. Statistically significant differences ($p < 0.05$), as determined by Tukey's HSD test, were indicated by bolded p-values. Lowercase letters indicate statistically significant differences ($p < 0.05$) between the treatment means.

	EC (μS/cm)	pH	Plant available nutrient concentration in soil (mg/kg)									Total (g/kg)	
			Ca	P	K	Mg	S	B	Cu	Mn	Zn	N	C
Biochar(ton/ha)													
B0	1.51	5.94 a	1798.94	13.96	159.95	203.08 a	21.37	0.78	15.27	20.38	8.02	2.40 a	35.09 a
B5+15	1.46	6.03 b	1794.52	13.73	157.76	207.78 ab	20.25	0.79	14.30	20.61	7.09	2.55 b	46.61 b
B10	1.54	6.01 ab	1836.03	13.86	153.34	212.33 b	21.50	0.77	19.11	18.95	7.00	2.40 a	35.58 a
SEM	0.05	0.03	34.28	0.20	3.84	2.28	0.75	0.02	3.58	1.35	0.46	0.03	1.85
Fertilizer rate, %													
F30	1.52	6.03	1857.18	12.71 a	141.63 a	203.53 a	18.25 a	0.72 a	14.13	15.32 a	6.23 a	2.43	38.95
F65	1.47	6.00	1773.26	13.70 b	155.00 b	206.43 a	20.21 b	0.79 b	19.11	18.68 a	7.43 b	2.46	39.37
F100	1.52	5.96	1799.05	15.14 c	174.42 c	213.23 b	24.65 c	0.83 b	15.44	25.95 b	8.44 b	2.45	38.96
SEM	0.05	0.03	34.28	0.20	3.84	2.28	0.75	0.02	3.58	1.35	0.46	0.03	1.51
p-values													
B	0.267	0.094	0.704	0.610	0.431	0.364	0.618	0.824	0.474	0.224	0.162	0.082	0.003
F	0.499	0.120	0.067	<0.001	<0.001	0.001	<0.001	<0.001	0.375	<0.001	0.001	0.694	0.952
B×F	0.579	0.491	0.426	0.565	0.288	0.512	0.453	0.029	0.323	0.515	0.055	0.128	1.000
Partial η2 values													
B	0.356	0.546	0.111	0.152	0.245	0.286	0.148	0.062	0.220	0.392	0.455	0.566	0.864
F	0.074	0.210	0.259	0.891	0.804	0.515	0.810	0.646	0.103	0.781	0.560	0.040	0.006
B×F	0.141	0.165	0.184	0.144	0.232	0.159	0.176	0.434	0.218	0.158	0.386	0.315	0.003

Biochar application rates are denoted as B0, B5+15, and B10, corresponding to 0, 5+15, and 10 tons per hectare, respectively. Fertilizer application rates are denoted as F30, F65, and F100, representing 30%, 65%, and 100% of the recommended dose, respectively. Ca = calcium, P = phosphorus, K = potassium, Mg= magnesium, S= sulfur, B= boron, Mn=manganese, Zn=zinc, N= nitrogen and C= carbon. SEM = standard error of the mean

5.3 Measurement of yield components of faba bean

Biochar application had a significant effect on the leaves weight ($p = 0.011$), stems weight ($p = 0.028$), vegetative mass ($p = 0.017$), 1 and 1000 seed weight ($p = 0.033$) (Table 4). Application of maximum biochar (5+15 ton/ha) increased leaves weight by 173.53% and vegetative mass by 105.84% compared to control treatment (no biochar). 10ton/ha biochar application showed the increase of stems weight by 83.32%, 1seed weight by 11.43% and 1000 seed weight by 9.31% than the control treatment, but the biochar treatments did not differ from each other statistically significantly. The highest AGB was recorded under 5+15 ton/ha biochar treatment, which increased by 73.44% in comparison to control treatment (Table 5). However, biochar treatment 10 ton/ha did not differ significantly from 5+15 ton/ha treatment.

Fertilizer application had a significant effect only on the number of side tillers/m² ($p = 0.019$), being statistically higher by 9.10% with application levels of 100% compared to the level of 30%. The yield in maximum biochar treatment was 46.51% higher compared to the control, although the difference was not statistically significant. The fertilizer treatments also did not have statistically significant effect on the yield, although 100% application level showed slightly higher value than the fertilizer levels of 30 and 65%. Harvest Index (HI) ranged increased from 0.48 to 0.53 while 100% fertilizer rate was applied, but there were no statistically significant differences between the fertilizer treatments were observed.

Interaction effects between biochar and fertilizer (BC×F) were not statistically significant, although a near-significant trend was observed for the number of seeds/pod ($p = 0.064$). The effect size was large ($\eta^2 > 0.14$) in application of biochar to all of the yield components except number of pod/stem and number of seed/pod. Fertilizer had large effect size only to the side tiller/ m² and leaves weight g/m² but there were large effect size for the interaction of biochar and fertilizer (BC×F) except in case of stems/ m², stems weight, AGB, pod/stems and the yield (ton/ha) (Table 5).

Table 5: Effects of biochar, fertilizer and their interaction on the yield components of faba bean. The values represented the means of 4 replicates for each of the 3 levels of treatment. Statistically significant differences ($p < 0.05$), as determined by Tukey's HSD test, were indicated by bolded p-values. Lowercase letters indicate significant differences ($p < 0.05$) of treatment means.

	Plant s densi ty /m ²	Main tiller /m ²	Side tiller/m ²	Stems /m ²	Leaves weight g/m ²	Stems weight g/m ²	Vegetative mass g/m ²	AGB (ton /ha)	pod/ stem	seeds/p od	1 seed weight, g	1000 seed weight, g	Yield (seed) ton/ha	HI
Biochar (ton/ha)														
B0	4.94	26.22	4.00	30.22	20.93 a	54.43 a	75.36 a	3.05 a	7.73	2.37	0.35	352.46	1.72	0.53
B5+15	5.94	37.33	8.89	46.22	57.25 b	97.88 b	155.12 b	5.29 b	6.53	2.30	0.35	347.49	2.52	0.46
B10	4.75	37.33	10.22	47.56	50.64 b	99.78 b	150.42 b	4.98 b	6.65	2.29	0.39	385.24	2.50	0.48
Fertilizer rate(%)														
F30	5.19	34.22	9.78b	44.00	53.38	90.76	144.14	4.48	6.88	2.23	0.37	367.37	2.16	0.48
F65	4.89	35.56	2.67 a	38.22	35.97	79.51	115.48	4.47	6.96	2.42	0.35	346.70	2.22	0.46
F100	5.56	31.11	10.67 b	41.78	39.48	81.81	121.29	4.38	7.06	2.31	0.37	371.12	2.35	0.53
p-value														
BC	0.250	0.264	0.100	0.106	0.011	0.028	0.017	0.070	0.710	0.897	0.033	0.033	0.425	0.403
F	0.570	0.863	0.019	0.809	0.207	0.690	0.356	0.987	0.992	0.436	0.319	0.313	0.930	0.507
BC × F	0.465	0.445	0.193	0.615	0.130	0.823	0.427	0.776	0.947	0.064	0.165	0.160	0.929	0.566
Partial η ² values														
B	0.370	0.358	0.536	0.527	0.777	0.698	0.742	0.588	0.108	0.035	0.680	0.679	0.248	0.262
F	0.061	0.016	0.356	0.023	0.160	0.040	0.108	0.001	0.001	0.088	0.119	0.121	0.008	0.073
BC × F	0.172	0.178	0.275	0.131	0.313	0.077	0.184	0.090	0.038	0.374	0.290	0.293	0.045	0.144

Biochar (BC) rates were denoted as B0, B5+15, and B10, corresponding to application rates of 0, 5+15, and 10 tons per hectare, respectively.

Fertilizer (F) applications were labeled F30, F65, and F100, indicating fertilizer rates of 30%, 65%, and 100% of the recommended dose.

6. Discussion

6.1 Nutrient enriched biochar and soil moisture content

The increase of soil moisture content with nutrient enriched activated biochar application corresponded with the previous study by Asai et al. (2009) in case of non-activated 16 ton/ha biochar application increased the soil water holding capacity which positively impacted plant water availability. This positive impact on soil moisture could be the reason of biochar properties like porosity, large surface area (Abel et al., 2013; Obia et al., 2016; Laird et al., 2010). Also Glaser et al. (2002) found that biochar added to soil at a rate of 45% of soil volume, increased water retention capacity upto 18% in sandy soil, which aligned with the result of this experiment as the experiment contained sandy clay loam soil. The application of biochar could improve the storage of soil moisture through improving the distribution of soil pore size that could lead to improve the soil water retention capacity (Novak et al. 2012 and Chen et al. 2010). So, biochar may improve soil aggregation, which in turn improves moisture retention as it is very important to maintain the soil moisture level above permanent wilting point; otherwise, plants can face water stress that may reduce the potentiality of targeted yield components.

As the control plots showed moisture levels that were close to the permanent wilting point (PWP), suggesting moisture stress that might restrict crop development. This observation was supported by the findings of Spokas et al. (2012), who demonstrated that biochar-amended soils performs better moisture retention and reduced water stress during key growth stages. Tammeorg et al., (2014a) found the similar findings of softwood biochar application in boreal condition on sandy clay loam soil where 5-20 ton/ha biochar was able to improve the water holding capacity. The special physicochemical characteristics of biochar, especially its porous structure, are responsible for the improved water retention in the soils that have been modified with it.

In hot and dry period of growing season, the biochar containing plots held water level clearly above the permanent wilting point compared to the control treatment, indicating that using of biochar may be an effective strategy for improving water availability. Temperature fluctuations also contribute to the soil moisture content through the rate of evaporation. During several warmer weeks (especially week 3, 7 and 11), the rate of evaporation from the soil had probably

been higher than during the weeks with lower temperatures. In that time, maximum biochar-treated soils enhanced water retention capacity to mitigate the water stress for the growing plants. For further studies the rate of evaporation can be measured to know precise effect solely on biochar as the addition of biochar can help in reducing water evaporation, particularly under warmer conditions (Yang et al., 2025).

6.2 Soil chemical properties

In this study, soil pH and concentration of plant available Mg were slightly increased by biochar addition, but there was no significant impact on soil chemical properties. This could be happened for several reasons as the impact of biochar on soil properties varies based on biochar's features and the soil to which it is applied (Joseph et al., 2021). The biochar effect on soil chemical properties like electrical conductivity, pH, content of easily soluble Ca, P, Mg, S and total N were not significantly ($p > 0.05$) affected by biochar application in earlier several years of this same experimental field which was conducted by Tammeorg et al., (2014a) . Amoah (2019) who explored the biochar effect on the soil chemical properties on same experiment. He found that there was no significant effect of biochar on soil chemical properties even after 9 years of biochar application. The lack of significant changes in these studies might be due to the relatively low application rates of biochar used and the high nutrient level of experimented soil originally.

Contrary to the results of this study, it has been reported that application of biochar has increased concentrations of P, K, N and Mg in soil (Laird., 2008 and Nelson *et al.*, 2011). The amount of BC addition may affect EC although in your own study, no effect was observed; however, in the study of Mia et al. (2014), EC increased 14-fold compared to the control treatment where the amount of biochar rate was many times greater than the amount used in our study,.

The slight increase in soil pH observed in this study as a results of biochar addition could have been due to the presence of alkaline metal oxides, hydroxides, and carbonates in biochar (Dai et al., 2017). The negatively charged functional groups, such as carboxyl, hydroxyl, and

phenolic groups on biochar, also bind H^+ ions from soil solution, reducing their activity and increasing soil pH (Chintala et al., 2013).

It's interesting to note that biochar did have a mild impact on the concentrations of magnesium (Mg) and zinc (Zn), with the B10 treatment slightly raising Ca, Mg and Cu concentration. This implies that biochar might have impacts on the availability of micronutrients, perhaps as a result of modifications to the microbial activity or cation exchange capacity (CEC) of the soil (Marschner, 2012). According to Kätterer et al. (2019), applying biochar enhanced the physical and chemical characteristics of the soil including pH, soil porosity, plant-available P, soil water-holding capacity, and nitrogen mineralization, and the grain yield of maize and soybeans in a ten-year study conducted in Kenya.

According to a number of studies, the benefits of biochar are frequently more noticeable in soils that are lacking in nutrients or that have poor structure (Lehmann & Joseph., 2015). The use of nutrient enriched biochar might not significantly change the pH or nutrient levels in well-fertilized soils, such the ones used in this study.

The fertilizer addition had no significant effect on soil pH or EC, even though the soil's nutrient levels clearly increased. As high fertilizer application rates can occasionally result in considerable soil acidification or salinization, the obtained results indicate that the fertilizer application rate in this study was well-balanced (Fageria et al., 2011).

Only boron (B) exhibited significant impacts from the biochar and fertilizer ($B \times F$) interaction, with a modest effect size. This suggests that the combination of biochar and fertilizer can affect the availability of specific soil nutrients. According to the fertilizer application rate, biochar may either increase or decrease the availability of particular nutrients, as indicated by the statistically significant interaction for boron ($p = 0.029$).

Nutrient enriched biochar increased total concentration of carbon which aligned with the finding of Sun et al. (2024) that at application rate of 0,5 and 10 ton/ha, soil carbon content increased. This impact mostly results from the inherent properties of biochar, which is considered an inert, carbon-rich aromatic organic material exhibiting high stability and resistance to biological

breakdown (Lehmann et al., 2011). Nutrient enriched biochar can increase the total soil carbon concentration that can assist carbon being retained in soil for a long time that may help to maintain soil fertility as well as improving soil carbon storage capacity.

Though application of activated biochar and fertilizer did not show any significant result on soil chemical properties (except on B concentration), the applied biochar with fertilizer did not exhibit any negative impact on soil at all. According to a previous study the effects of applying spruce chip biochar, either alone or in conjunction with inorganic fertilizer or meat and bone meal (MBM), to a nutrient-deficient loamy sand soil were assessed during a two-year field study carried out in a boreal environment. The results concluded that applying biochar had no detrimental effects on the physicochemical characteristics of the soil or on the quantity or quality of spring wheat produced (Tammeorg et al., 2014b). Further research is needed to investigate the long-term impacts of nutrient enriched biochar and fertilizer interactions on soil health, nutrient dynamics, and crop productivity under a variety of environmental circumstances.

6.3 Yield formation and yield components of faba bean

Activated biochar increased the weight of faba bean's leaves, stems and seeds. This is in accordance with the results of Vaccari et al. (2011), who found that biomass production and yield of durum wheat increased up to 30% by biochar addition. An organically activated biochar rate of 2–4 tonnes per acre was also recorded as highest growth improvement and attributed yield components of maize cultivation (Iqbal et al., 2024). In the early growing season measured plant density/m² had no impact from biochar but showed the highest in number where activated biochar was in the maximum biochar rate (5+15 ton/ha) but at the end of the season there was no significant impact was observed. In this experiment activated biochar did not show any significant effect on faba bean yield components though 1 and 1000 seed weight had significant increase at the moderate biochar rate(10 ton/ha) plot. These findings align with previous research indicating that biochar can enhance nutrient use efficiency and seed development (Biederman & Harpole., 2013). In comparison to no biochar treatments for lettuce and cabbage, Carter et al. (2013) reported overall increases in biomass, root biomass, plant height, and leaf count throughout all cropping cycles. A study by Kalu et al. (2021) found the similar result from the same experimental field of our study by applying similar amount of

softwood biochar on different leguminous crop that increased the crop biomass. N-enriched biochar application at a rate of 4ton/ha increased crop biomass that was found from the study of Hei et al. (2023).

The nutrient enriched biochar application produced the highest amount of vegetative mass and AGB, nearly doubling the amount compared to the control (B0) treatment. It suggests that applying activated biochar in early stages of growing may have a synergistic effect on biomass production, especially when it comes to promoting the growth of both leaves and stems. So, activated biochar may increase the vegetative mass as well as seeds weight. This enhancement can happen due to the improved soil water retention of our study.

The harvest index (HI) was highest in the control treatment, although not statistically significantly, indicating that while biochar increases biomass, the relative allocation to grain yield does not always rise correspondingly. Increased vegetative growth can sometimes lead to delayed reproductive development, especially if the plant invests too much energy into producing leaves and stems. This can result in lower seed production and, consequently, a lower harvest index. If the plant is "too healthy" in terms of its vegetative parts, it might not allocate enough energy to seed production at the right time in the growing season (Kozłowski & Wiegert, 1986) . There were lots of weed in the field during growing season that may compete (Horvath et al., 2023) with the growing crops for sunlight, water as well as nutrients to affect the overall growth as well as yield of faba bean.

The effect of fertilizer on faba bean yield components was somewhat limited. The increased number of side tiller/m² due to fertilizer application supports that optimal nutrient availability can promote tiller initiation (Marschner, 2012). However, fertilizer alone did not significantly affect yield parameters such as AGB or seed yield, indicating a restricted response of faba bean to increasing fertilizer rates in the absence of biochar. Partial η^2 values support this interpretation, showing mostly small effect sizes for fertilizer application. This may be due to the leguminous nature of faba bean, which fixes atmospheric nitrogen and may require less supplemental fertilization under favorable conditions (Jensen et al., 2010).

The weather condition was considerably warmer than earlier growing season and also compared to 30 years average that may affect the yield formation of faba bean through the proper utilization of biochar, fertilizer or their interaction of this experiment. As there were no statistically significant interactions between biochar and fertilizer, indicating that the effects of each factor are essentially independent under the conditions used in this study. Moderate interaction effects were noted for parameters such as leaves weight, seeds per pod and 1000-seed weight, indicating the possible synergistic responses that were not fully captured by the experimental design like application rate or sample size. Further investigation of activated biochar application on several crops to improve soil physical, chemical and biological properties will help to know the plants mechanism against activated biochar application and producing targeted yield profitably.

7. Conclusion

Nutrient enriched activated biochar application improve soil water holding capacity especially during crucial stages of faba bean development such as flowering and seed formation. This increase in moisture retention is especially beneficial in agricultural systems where water availability is restricted, since it helps to minimize the detrimental effects of drought and ensures more steady crop yields in dry conditions. Biochar addition positively influenced the stem and leaf biomass as well as vegetative mass and seeds weight of faba bean, highlighting biochar's ability to stimulate plant growth though biochar's impact on soil nutrient concentrations was not significant. While biochar application promoted vegetative growth, the harvest index (HI), which indicates the fraction of biomass aimed to grain yield, did not significantly increase. This phenomenon indicates the need for careful management of biochar application rates to avoid excessive biomass production despite the loss of grain yield. Further studies are required to gain a comprehensive understanding of the long-term impacts of biochar on soil health, nutrient cycling, and crop productivity across various environmental contexts. Biochar has the potentiality to improve water holding capacity that might be a sustainable way of crop cultivation.

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9. Use of artificial intelligence

Chat GPT has been used to check the grammar, spelling and as a proof reading

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